

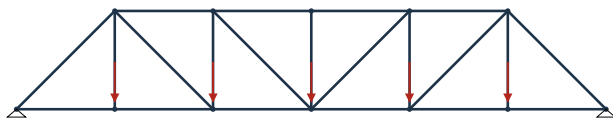
ReSolve: Parse, Patch, and Check Engineering SVGs

Anonymous CVPR submission

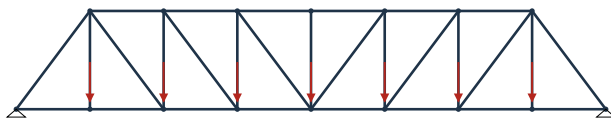
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PARSE + PATCH + RE-IMPORT + ENGINEERING CHECK

Source SVG

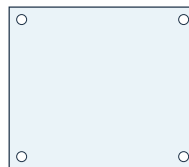


Edited SVG



Structural patch: 6 to 8 panels; 21 to 29 members.

FEM re-solve: peak stress 84.38 $\rightarrow$ 112.50 MPa; equilibrium passes.



Increase hole diameters by 6 mm: edit is faithful; clearance fails.

Recovered edge distance: 15 $\rightarrow$ 12 mm, below the stated 15 mm minimum.

Figure 1. Two edits, re-imported and re-checked. Top: “Change the bridge to 8 panels” on a Pratt truss with span $L=12$ m, depth $h=2$ m and member area $A=800$ mm² ($E=200$ GPa). The 53-operation patch repositions 36 existing elements, inserts 8 members, 4 joints and 2 load arrows, and updates the title, giving 29 members. The drawing specifies $P=15$ kN per loaded node, so the loaded bottom-chord nodes increase from 5 to 7 (total load 75 $\rightarrow$ 105 kN). Axial FEM on the re-imported drawing gives a peak member stress of 84.4 $\rightarrow$ 112.5 MPa, equal to the midspan chord force $PLn/(8h)$ divided by A for $n=6 \rightarrow 8$ panels, and a peak displacement of 8.95 $\rightarrow$ 11.85 mm; the relative equilibrium residual is below 10^{-13} . This confirms that the edited truss is solvable and in equilibrium. No stress limit is applied, so it does not show that the bridge is safe. Bottom: “Increase all hole diameters by 6 mm” on a 300 $\times$ 260 $\times$ 9 mm plate whose hole centres lie $c=20$ mm from both adjacent edges. The two-operation patch changes d from 10 to 16 mm and updates the hole note, matching the target SVG tree exactly. The clear edge distance $e=c-d/2$ falls from 15 to 12 mm, below the 15 mm minimum. The source holes already sit at that minimum, so with fixed centres the rule allows only $d \leq 2(c-15) = 10$ mm; 16 mm holes need $c \geq 23$ mm. The request is infeasible as stated: the editor applies it without moving the holes, and the check reports the violation. This is a geometric clearance rule, not FEM. Geometry is cropped from the source and edited SVGs, with labels omitted and colors normalized.

Abstract

001 We study engineering SVG editing through source-tree
002 parsing, parsed and validated patches, and physical
003 or geometric checks on the recovered output. The
004 evaluated implementations cover electrical and hy-

draulic networks, axial trusses and perforated plates.
A learned 1.5B network editor reaches 100 % on 1,000
held-out edits and 99.5 % on larger networks under
identifier addressing, paired with circuit and hydraulic
solvers. Separately, a bounded deterministic truss/-
plate editor reproduces all 40,000 targets in a saved

005
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011 pipeline audit and flags 699 faithful plate edits that vio-
012 late clearance rules. Worked SVG examples show mem-
013 ber resizing, load changes, structural topology patches
014 and hole enlargement. Axial FEM recomputes stress,
015 displacement and equilibrium; geometric checks mea-
016 sure edge distance and ligament. Saved Astra cases
017 expose a false pipe-network verdict and repeated re-
018 fusals to analyse a correctly edited 21-member frame.
019 Results retain the input contract and checker of each
020 evaluated editor.

021 1. Introduction

022 Editing an engineering drawing requires several linked
023 updates. Moving a column changes member lengths;
024 changing a bridge’s panel count changes connectivity
025 and loading; enlarging holes changes clearances; and
026 modifying a pipe redistributes flow. A useful SVG ed-
027 itor must preserve the intended design, update its ge-
028 ometry and annotations, and recompute the quantities
029 affected by the edit. Figure 1 shows two implemented
030 examples: a bridge edit followed by FEM and a faithful
031 plate edit whose reduced edge distance is flagged by a
032 geometric check.

033 Markup or pixel similarity alone cannot establish
034 these properties. SVG-editing benchmarks [14, 17]
035 measure output similarity, while engineering-reasoning
036 benchmarks [5, 11, 12] evaluate numerical answers or
037 simulation tasks. Our question concerns the delivered
038 artifact: does the edited SVG describe the requested
039 design, and what happens when that design is anal-
040 ysed under the stated assumptions?

041 We use an edit–recover–check workflow. The editor
042 parses the source into a scene tree, predicts or con-
043 structs a patch, parses and validates that patch, and
044 applies it to the original SVG. Structural operations
045 insert nodes when topology changes. A family-specific
046 importer then recovers the produced drawing. The
047 applicable solver or geometry checker evaluates that
048 recovered result. Edit fidelity and engineering accep-
049 tance are reported separately: a correct edit can intro-
050 duce a violation, and passing one check does not certify
051 every aspect of a design.

052 The network implementation, ReSolve, recovers vis-
053 ible wire and pipe connectivity, component symbols
054 and labels without identifiers or hidden metadata. A
055 learned patch editor supplies the edits; modified nodal
056 analysis and hydraulic flow–head equations supply the
057 verdicts. The structural/mechanical extension uses
058 bounded edit interpretation and importers for trusses
059 and plates, followed by axial FEM or manufacturing-
060 geometry checks. A separate frame harness uses a
061 supplied physical model and checks returned member

geometry before a planar-frame FEM solve. We re- 062
report these contracts explicitly rather than pooling their 063
scores into a single cross-domain accuracy. 064

Contributions. 065

- 066 • Editing followed by verification of the resulting 067
SVG. The supported routes cover circuits, pipe net- 068
works, axial trusses and perforated plates. Worked 069
examples expose the requested revision, changed 070
geometry and recomputed engineering quantities 071
(Sec. 6).
- 072 • A learned network editor with a drawing-grounded 073
verifier. The 1.5B identifier-addressed editor 074
achieves 100 % on 1,000 held-out network edits and 075
99.5 % on larger networks. The verifier passes 10 076
analytical controls and recovers 10,000 generated 077
drawings exactly (Secs. 3 and 5).
- 078 • Reproducible datasets and checked edit examples. 079
We provide 10,000 network edits, a 400-edit harder 080
network tier, and a saved 40,000-edit determinis- 081
tic truss/plate pipeline audit. The latter identifies 082
699 requested plate edits that are faithful but vio- 083
late clearance rules. These are pipeline results, not 084
learned-editor accuracy (Secs. 4 and 6).
- 085 • Audited analysis failures and limitations. The saved 086
Astra pipe error, repeated frame analysis refusal, 087
opaque-identifier control and 100-case cross-domain 088
learned-patcher pilot show where editing, interpre- 089
tation and checking must be evaluated indepen- 090
dently.

091 2. Related work

092 Vector graphics generation and editing. Learned SVG 093
generators model paths and primitives directly [1] or 094
emit SVG code with a language-model decoder [15]. 095
Instruction-based SVG editing is benchmarked by com- 096
paring the edited result with a reference image or 097
markup [14], and scientific-diagram work learns edits 098
from paper revisions [17] or targets structure-faithful 099
generation [10]. These methods treat a drawing as an 100
image or a document. We treat an engineering drawing 101
as a specification of a physical network and score it by 102
what that network does.

103 CAD and engineering design. Generative CAD mod- 104
els produce parametric construction sequences [9, 19], 105
and agentic systems couple language models to sim- 106
ulators for mechanics [13] and finite-element analy- 107
sis [4, 20]. Benchmarks such as FEABench [12], FEM- 108
Bench [11] and PDEAgent-Bench [5] evaluate the rea- 109
soning or code that sets up a simulation. Our verifier 110
instead reads a finished two-dimensional drawing and

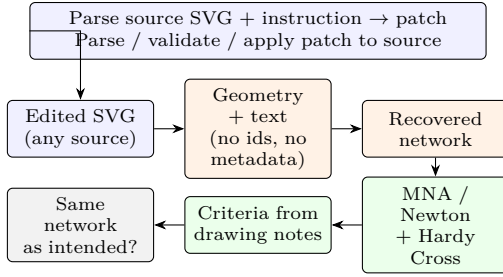


Figure 2. The network implementation of ReSolve. An edited network drawing is read as built: geometry and labels become a network, the network is solved, and the result is checked against the criteria printed on the drawing and compared with the intended design. The network importer does not read identifiers or metadata.

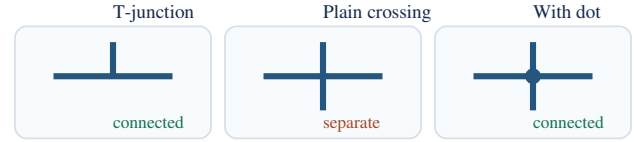


Figure 3. Connection conventions read from visible geometry. A segment ending on another makes a T-junction. Two segments that cross remain separate unless a filled dot marks their intersection. The same rules apply to wires and pipes.

3.1. Recovering a network from a drawing

Recovery parses the SVG with transforms applied and rejects scripts, external references, clipping and masking. It then reads only what is visibly drawn.

Connectivity. Stroked straight segments are candidate conductors (for pipe networks, only heavy strokes). Figure 3 shows the connection rules. A segment endpoint lying in the interior of another segment is a T-junction and splits it; a filled junction dot splits every segment passing through its centre. Two segments that merely cross stay unconnected, which is the drafting convention for both schematics and piping plans. Endpoints closer than $\tau = 10^{-3}$ drawing units are merged. For circuits, all points joined by wire form one electrical node. For pipe networks, nodes are points of degree $\neq 2$, dotted points and points on a reservoir outline, and every chain of segments between two nodes is one pipe. A 90° turn along a chain is an elbow; other bend angles are rejected.

Components. A rectangle whose two short-side mid-points coincide with conductor endpoints is a resistor between the corresponding nodes. A circle is classified by where conductors meet it, not by its size: if a conductor passes through its centre it is a junction dot, and if exactly two conductors end on its rim it is a voltage source, whose positive terminal is the one nearer the “+” mark inside it. Filled rectangles touched by pipe ends are reservoirs. Classifying by incidence rather than radius or stroke width keeps recovery unchanged under uniform scaling of the drawing.

Values and criteria. Labels are parsed with fixed grammars (e.g. R3 4.7 k Ω , P5 320 m $\varnothing$ 150, J2 q=6 L/s z=10 m) and assigned one-to-one to the nearest component or pipe, within a reach proportional to the drawing’s own scale (the median resistor body or pipe segment). Design criteria are read from the drawing’s notes: the resistor power rating and supply fuse; the

reconstructs the model it implies, so it can audit the artefact a designer or a model actually delivers.

Network analysis. Modified nodal analysis [6] is the standard formulation for linear circuits. Pipe networks are classically solved by loop corrections [3] or by the gradient method on flows and heads [18] that underlies EPANET [16]; we use Churchill’s friction factor [2], which is continuous from laminar through fully rough flow. None of this is new. What is new is running these solvers on networks recovered from free-form drawings, and using them as the metric for text-guided editing.

3. Network editing and drawing-grounded verification

Patch-based editing. The editor receives the parsed source SVG and an instruction and emits operations on that source tree. A patch parser checks the operation schema; policy validation checks targets and allowed mutations before execution. The executor serializes the updated tree as the output SVG. The engineering importer then reads this executed artifact for verification. The full-SVG outputs in the saved Astra experiment are a separately identified baseline; they are not the editing interface of the patch model.

Task. A source drawing S of a network and an instruction u (e.g. “Add $R7 = 10\Omega$ in parallel with $R1$ ”) define a target design D^* . An editor returns a drawing $\hat{S}$. We ask two questions of $\hat{S}$: does it describe D^* (edit correctness), and does the network it describes satisfy the criteria stated on the drawing (verdict)? Both are decided by recovering a network $\mathcal{R}(\hat{S})$ from the drawing and solving it.

pipe roughness, elbow loss coefficient, minimum pressure head and maximum velocity. A drawing that leaves any component unlabelled, or has two labels competing for one component, is reported as unreadable rather than guessed at.

3.2. Solving the recovered network

Circuits. We use modified nodal analysis [6] with one grounded node per connected part of the network, so a floating fragment carries no current and does not make the system singular. A source whose terminals share a node, or a loop made only of sources, makes the system singular; such a drawing is reported as ill-posed. The outputs are every branch current, every resistor’s dissipation and every source’s current; the criteria are $P_R \leq P_{\text{rated}}$ for each resistor and $|I_V| \leq I_{\text{fuse}}$ for each source.

Pipe networks. For pipe p from node a to node b with flow Q_p , the head loss is

$$h_p(Q_p) = \left(f(\text{Re}_p, \varepsilon/D_p) \frac{L_p}{D_p} + n_p K_{\text{elbow}} \right) \frac{v_p |v_p|}{2g}, \quad (1)$$

where $v_p = 4Q_p/(\pi D_p^2)$, f is Churchill’s friction factor [2] and n_p is the number of elbows counted from the drawing. We solve jointly for pipe flows and junction heads,

$$h_p(Q_p) = H_a - H_b \quad \forall p, \quad \sum_{\text{in}} Q_p - \sum_{\text{out}} Q_p = q_j \quad \forall j, \quad (2)$$

with reservoir heads fixed, by Newton’s method with backtracking [18], stopping at residuals below 10^{-10} m and 10^{-13} m³/s. Because Churchill’s f tends to $64/\text{Re}$ in laminar flow, h_p has a finite, non-zero slope at $Q_p = 0$ and the Jacobian stays regular when a flow reverses. As an independent check, every network is also solved by Hardy Cross loop corrections [3], with multiple reservoirs joined through a virtual datum node. The criteria are $|v_p| \leq v_{\text{max}}$ for each pipe and $H_j - z_j \geq p_{\text{min}}$ for each junction.

3.3. Scoring an edited drawing

An edit is correct when $\mathcal{R}(\hat{S})$ equals the target network D^* in two senses. The first is topological: we compare, for every node, the multiset of named component terminals it touches, which is a complete invariant of a network with named components, together with every component value and every criterion. The second is physical: the two solutions agree, with currents within 10^{-9} relative plus 1 pA, flows within 10^{-9} m³/s and heads within 10^{-7} m. Markup is irrelevant: a drawing that re-orders elements, renames identifiers or re-formats numbers is credited if it describes the same

Control	ReSolve	Reference
series current $V/(R_1+R_2)$	0.0375	0.0375
parallel current $V(1/R_1+1/R_2)$	0.174545	0.174545
balanced Wheatstone bridge current	-1.35×10^{-18}	0
unbalanced bridge vs Thevenin	-0.00307525	-0.00307525
superposition of two sources	0.263636	0.263636
two reservoirs vs bisection (m ³ /s)	0.0324668	0.0324668
laminar loss vs Hagen-Poiseuille (m)	4.17×10^{-5}	4.17×10^{-5}
Churchill vs Colebrook at $\text{Re}=1\text{e}6$ (relative gap)	1.00497	1
Newton vs Hardy Cross, two loops + two reservoirs (m ³ /s)	1.22×10^{-14}	0
mass balance: reservoir supply = demand (m ³ /s)	1.39×10^{-17}	0

Table 1. Analytical controls for the two solvers. References are closed forms, an independent bisection, or a second algorithm.

network, and a drawing that is byte-for-byte plausible but disconnects a wire by 0.01 units is not.

3.4. Validating the verifier

Table 1 lists the analytical controls. They include a balanced and an unbalanced Wheatstone bridge; we initially chose bridge values that happened to balance, which made the “unbalanced” control vacuous and hid a sign error, and we record this as a caution. Beyond the controls, 10,000 independently generated drawings (source designs and all their edits, over two seeds) recover to exactly the network they were drawn from, as do all 20,800 drawings in the released datasets. Unit tests confirm four further properties: recovery is unchanged by stripping identifiers, shuffling elements, translating and uniformly scaling the drawing by $\times 0.25$ to $\times 4$; a T-junction joins, while a crossing joins only when dotted; moving only the “+” mark reverses every branch current; and moving one wire endpoint by 0.01 units breaks the connection. Two of these tests found real defects in the first version. A fixed-radius rule for junction dots misclassified dots once the drawing was scaled up, and an absolute current tolerance misjudged an edit whose two sources exactly cancel.

4. Network datasets

Resistor networks. A design starts from a grid of $2 \times 3 \times 3$ nodes. Edges are removed at random while the graph stays connected and bridgeless (a bridge would carry no current), until 2–6 independent cycles remain. Each edge becomes a resistor (E6 series, 10Ω – $6.8 \text{ k}\Omega$), a plain wire (20%), or one of one or two DC sources (5–48 V, random polarity). Designs are rejected if a resistor is shorted, if sources form a loop, or if no current

flows. Notes state a resistor power rating (0.25–2 W) and a supply fuse (0.1–1 A). Drawings use standard symbols, with junction dots at nodes of degree three or more.

Water-distribution networks. A random spanning tree of a $2\text{--}3 \times 3\text{--}5$ grid receives one to three extra pipes to close loops. Some dead ends are pruned, and one or two reservoirs attach at the boundary. Grid points of degree $\neq 2$, plus about a third of the rest, become junctions carrying a demand (0–15 L/s) and an elevation (0–30 m). The remaining pass-through points become straight runs or 90° elbows. Pipes have lengths of 100–800 m and nominal diameters of 80–300 mm, and notes state the roughness, elbow loss coefficient ($K = 0.9$), minimum pressure head (10–20 m) and maximum velocity (1.5–2.5 m/s).

Edits. Each design receives four instructions. For circuits these are a resistor value change, a source voltage change, a polarity reversal (two-source designs) or opening a branch (one-source designs, provided the network stays solvable), and adding a resistor in parallel with an existing one, drawn as a tapped branch. For pipe networks they are a diameter change, a demand change, removing a pipe (provided every junction stays supplied and no reservoir is cut off), and adding a pipe between adjacent junctions (or, if none exist, a length change). Target drawings are rendered from the edited design. The gold edit is a tree patch derived between source and target SVG, applied with a validating executor and required to reproduce the target exactly. A row is kept only if both drawings recover to their designs and, for pipes, Newton and Hardy Cross agree to $10^{-9} \text{ m}^3/\text{s}$.

Statistics. Table 2 summarises both tiers. The hard tier uses $4\text{--}5 \times 5\text{--}6$ grids with 8–16 independent loops for circuits and four to eight added pipes for water networks. It is evaluation-only: no model is trained on networks of that size. Splits are by source design, so the four edits of one drawing never cross splits.

5. Learned network-editor evaluation

Astra failure probe. We evaluate Astra (`gpt-6-astra`) through the Chat Completions API at medium reasoning effort, with a 32,000-token completion cap, no tools, `store=false` and one sample per task. Astra works in its natural mode: it receives the drafting conventions, the physical model (fluid properties, friction law, loss conventions) and the criteria in the prompt, and it returns the complete edited SVG followed by a JSON

	Main (v1)	Hard
Source designs	2,500	100
Edits (train/val/test)	8,000/1,000/1,000	0/0/400
Circuit loops	3 (1–6)	11 (8–16)
Pipe-network loops	2 (1–4)	7 (4–9)
Edited designs safe	5,368	118
Verdict flips	780	22
Edits using <code>set_text</code>	5,722	209
Edits using <code>set_attributes</code>	490	18
Edits using <code>remove_element</code>	2,009	82
Edits using <code>insert_subtree</code>	1,779	91

Table 2. Datasets. Loops are independent cycles of the recovered network (median, range). “Verdict flips” are edits that turn a safe design unsafe or the reverse. All patches reproduce their targets and all drawings recover to their designs.

analysis stating the verdict, the violating components and the governing quantities. The edited SVG is scored by ReSolve exactly as any other drawing (Sec. 3), and the analysis is compared with the solver’s. Tasks are the first ten test edits by identifier for every (family, edit type) pair of the main tier (100 tasks), and the first four of the hard tier (40 tasks). The API account ran out of credit during the run. All 79 main-tier and 23 hard-tier tasks that completed are reported. Every circuit task completed. The missing tasks are all pipe networks, which failed with “no credits remaining” (26) or a dropped connection (12), and they are excluded rather than counted as failures.

Our editor. ReSolve pairs the verifier with a small editor: Qwen2.5-Coder-Instruct [8] at 1.5B and 7B parameters, fine-tuned with LoRA [7] (rank 16 on the attention projections) for two epochs on the 10,000-edit training split to map source SVG + instruction to a tree patch. The patch is applied by a validating executor, and the resulting drawing is recovered and solved, so the verdict is computed, not generated. Decoding is greedy. The hard tier is never seen in training. We compare two ways for a patch to name the elements it changes. Positional addressing uses the executor’s native node identifiers, where `n77` is the 77th element in document order. Id addressing uses the element’s own SVG identifier (`#R5-label`) and writes insertion at the end of a group as `"end"`. It is translated to positional form before execution, so the two differ only in what the model must write.

Metrics. Edit: the edited drawing recovers to the target network. Verdict: the stated pass/fail matches the solver’s. Violations: the set of violating components matches exactly. All: all three at once. For systems

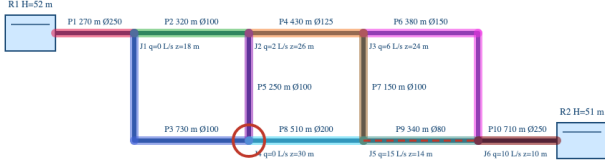


Figure 4. Astra’s wrong verdict on its edited SVG. “Change the length of P9 to 340 m” (dashed) in a two-reservoir network with four loops. Astra’s edit is correct, and it reports the minimum pressure head at J4 (circled) as 20.04 m and the design as passing. Re-solving the drawing gives 19.23 m, below the 20 m minimum stated on the drawing. Several of its pipe velocities are off by up to 67 %.

whose verdict is computed by ReSolve, the verdict is taken from the solve of the edited drawing, whether or not the edit was correct.

5.1. Control cases and failure context

Astra is an accurate editor of these drawings. Every one of its completed edits recovers to the target network, including parallel branches drawn as tapped sub-circuits and pipes added between junctions, and it never disconnected a wire or broke the drafting conventions. Its circuit analysis is exact on every completed task, including hard-tier networks with up to 16 independent loops. We did not find the failure boundary we expected for linear networks: at these sizes, unaided nodal analysis is within reach. Each answer costs \$0.20 on the main tier and \$0.38 on the hard tier on average, and takes a median of 44 s and 101 s respectively, with no truncated responses.

5.2. Astra’s silent wrong verdict

The only substantive error among Astra’s completed network answers is in a nonlinear pipe network, and it is not flagged (Fig. 4). The edit is right, but the analysis declares a failing design safe: it reports 20.04 m of pressure head where the drawing yields 19.23 m against a 20 m minimum, and its flows are wrong throughout the loop that contains the edited pipe. Nothing in the answer distinguishes it from the correct ones: it finished normally, used a similar reasoning budget, and presented its numbers with the same precision. This is the failure mode a verifier exists for. It is rare (1 of 79 main-tier answers), it concentrates where the physics is nonlinear and the design sits near a limit, and it cannot be detected from the answer itself. Replacing Astra’s analysis with the solve of its own drawing removes it (Tab. 3, second row). The source drawing already puts J4 below the 20 m limit (19.26 m); the P9 edit leaves it below that limit (19.23 m). Figure 6 shows both drawings and the saved answer in detail.

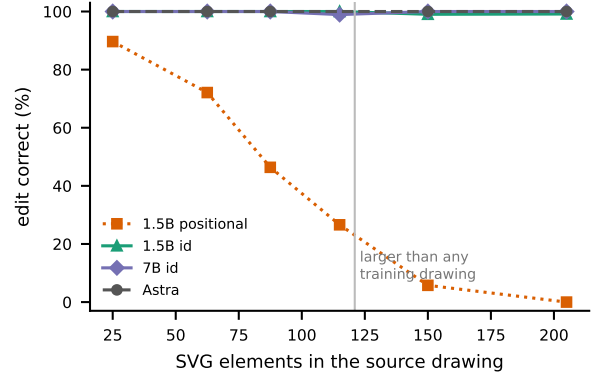


Figure 5. Edit accuracy against drawing size, pooled over both tiers and binned by the number of SVG elements in the source drawing (bins with at least five edits). The grey line marks the largest drawing in the training split.

5.3. The small editor

Table 4 and Fig. 5 show the editor’s results. With id addressing, the 1.5B editor is correct on every one of the 1,000 main-tier test edits, and on 99.5 % of the hard tier, whose networks are larger than anything in training. On the tasks Astra completed it matches Astra’s edits exactly (Tab. 3). Because its verdict is computed from its own drawing, its verdicts are correct whenever its edits are. A 7B editor trained identically reaches the same totals (1,000 of 1,000 and 99.5 %) with different misses (below). At this task size, model scale buys nothing that the representation does not already provide.

Addressing matters more than scale. The same model and data with positional addressing reaches only 67.5 % (circuits 43.6 %, pipes 91.4 %) and collapses to 16.5 % on the hard tier. Accuracy falls steadily with the number of elements in the drawing (Fig. 5): to name a label as `n77` the model must count through the document. In pipe drawings a pipe’s label position follows from its number and the pipe count, while in circuits it depends on how many sources precede it (each contributes three text elements), and circuits are where positional addressing fails. Naming elements by their SVG identifier removes the counting and costs nothing at inference.

Opaque-id control. This advantage depends on how identifiers are named. We replaced every SVG identifier in each hard-tier source and target with a unique, opaque token, while keeping visible geometry, labels, instructions and target physics unchanged. All 400

System	Edit	Verdict	Violations	All
Main tier (same completed tasks for every system)				
Circuits				
Astra (own analysis)	100 (50/50)	100 (50/50)	100 (50/50)	100 (50/50)
Astra drawing + ReSolve verdict	100 (50/50)	100 (50/50)	100 (50/50)	100 (50/50)
Qwen2.5-Coder 1.5B, untrained	0.0 (0/50)	0.0 (0/50)	0.0 (0/50)	0.0 (0/50)
ReSolve editor 1.5B	100 (50/50)	100 (50/50)	100 (50/50)	100 (50/50)
ReSolve editor 7B	100 (50/50)	100 (50/50)	100 (50/50)	100 (50/50)
Pipes				
Astra (own analysis)	100 (29/29)	96.6 (28/29)	96.6 (28/29)	96.6 (28/29)
Astra drawing + ReSolve verdict	100 (29/29)	100 (29/29)	100 (29/29)	100 (29/29)
Qwen2.5-Coder 1.5B, untrained	0.0 (0/29)	0.0 (0/29)	0.0 (0/29)	0.0 (0/29)
ReSolve editor 1.5B	100 (29/29)	100 (29/29)	100 (29/29)	100 (29/29)
ReSolve editor 7B	100 (29/29)	100 (29/29)	100 (29/29)	100 (29/29)
Hard tier (8–16 loops for circuits; unseen in training)				
Circuits				
Astra (own analysis)	100 (20/20)	100 (20/20)	100 (20/20)	100 (20/20)
Astra drawing + ReSolve verdict	100 (20/20)	100 (20/20)	100 (20/20)	100 (20/20)
ReSolve editor 1.5B	100 (20/20)	100 (20/20)	100 (20/20)	100 (20/20)
ReSolve editor 7B	100 (20/20)	100 (20/20)	100 (20/20)	100 (20/20)
Pipes				
Astra (own analysis)	100 (3/3)	100 (3/3)	100 (3/3)	100 (3/3)
Astra drawing + ReSolve verdict	100 (3/3)	100 (3/3)	100 (3/3)	100 (3/3)
ReSolve editor 1.5B	100 (3/3)	100 (3/3)	100 (3/3)	100 (3/3)
ReSolve editor 7B	100 (3/3)	100 (3/3)	100 (3/3)	100 (3/3)

Table 3. Edit and verdict accuracy, % (count), on the tasks Astra completed. “Astra drawing + ReSolve verdict” keeps Astra’s edited SVG but replaces its analysis with the solve of that drawing.

re-derived gold patches still reproduce and solve their targets. Without retraining, the saved 7B editor falls from 398/400 correct edits to 154/400 (67/200 circuits and 87/200 pipe networks). This is a limit of the editor trained on informative IDs; the verifier remains identifier-invariant.

Remaining errors and markup. The 1.5B editor’s two hard-tier misses both add a parallel branch to the largest circuits: the model misplaces the branch’s vertical coordinates by 8–30 units, so it no longer taps the intended wires, on rows lower in the drawing than any seen in training. The 7B editor gets every hard circuit right. Its two misses add a pipe to the largest water networks, drawing the pipe exactly but placing its label 91–100 units below it, which leaves the new pipe unlabelled. The verifier rejects all four drawings, the first two because the recovered network differs and the last two because they cannot be read. In the other direction, 22 of the 1.5B editor’s main-tier edits (6 of the 7B’s) are correct networks but not byte-identical to the gold drawing. For the 1.5B editor, 18 leave a junction dot that the gold drawing removes after opening a branch, and 4 place an added pipe’s label 1–12 units from the gold position. A markup metric would count these as failures. Scoring by the recovered network credits them.

Editor	Circuits	Pipes	All
1.5B pos., main	43.6 (218/500)	91.4 (457/500)	67.5 (675/1000)
1.5B pos., hard	0.5 (1/200)	32.5 (65/200)	16.5 (66/400)
1.5B id, main	100 (500/500)	100 (500/500)	100 (1000/1000)
1.5B id, hard	99.0 (198/200)	100 (200/200)	99.5 (398/400)
7B id, main	100 (500/500)	100 (500/500)	100 (1000/1000)
7B id, hard	100 (200/200)	99.0 (198/200)	99.5 (398/400)

Table 4. Our editor on the full test sets: edit correctness by family and end-to-end accuracy (“pos.” = positional, “id” = id addressing) on all 1,000 main-tier and 400 hard-tier edits, % (count).

5.4. Cost of a verified edit

An Astra answer costs \$0.20 (main tier) to \$0.38 (hard tier) at list price and takes a median of 44s to 101s. Our 1.5B editor trains in 36 minutes on one RTX PRO 6000 GPU and then needs 1.11s of GPU time per edit when batched sixteen at a time. The id-addressed run shared the GPU with another user’s jobs, which stretched its training to 90 minutes and inference to 1.53s per edit. The 7B editor trains in 111 minutes on one H100 NVL and needs 1.30s per edit. The verifier needs a median of 4ms (main) and 13ms (hard) of laptop CPU time per drawing to recover, solve, check and compare it. It is cheap enough to run on every edit from any source, including Astra’s.

447 6. Structural and mechanical SVG editing

448 The engineering workflow extends beyond the network
449 experiments through separate, bounded implementa-
450 tions. Table 5 states which editor and checker are ac-
451 tually evaluated. This distinction matters: the learned
452 network model has not been evaluated as a general
453 frame or truss editor.

454 6.1. Re-importing the edited structure

455 The truss/plate route recovers an engineering repre-
456 sentation from supported untagged SVG geometry and
457 visible annotations. A bounded request interpreter ap-
458 plies arithmetic and named parameter changes. For
459 fixed topology, the resulting target is converted to a
460 tree patch, serialized, parsed, policy-checked and ap-
461 plied to the original SVG. The actual patched SVG is
462 imported again before verification. Changing the panel
463 count uses version-3 structural patches with subtree in-
464 sersion and geometry replacement. The bounded plan-
465 ner builds an internal target representation for patch
466 derivation; the delivered SVG is always obtained by
467 applying the serialized, parsed and validated patch to
468 the original source. The 21-to-29-member bridge ex-
469 ample uses 53 operations: one text edit, 32 attribute
470 edits, four geometry replacements and 16 subtree in-
471 sions. Re-import and FEM check the executed result.
472 This is a deterministic patch compiler, not an evaluated
473 learned topology-patch predictor.

474 A concrete parsed patch. In the section-depth exam-
475 ple, the source parser assigns the section annotation
476 the scene address n34. The serialized edit is:

```
477 {"version":2,"operations":[  
478   {"op":"set_text","targets":["n34"],  
479    "text":"Section: 30 x 40 mm"}]}
```

480 The parser resolves this address against the source tree;
481 the executor changes that annotation and preserves the
482 remaining elements. Re-importing the output recov-
483 ers the larger axial area and FEM reports peak stress
484 5.930 MPa instead of 11.861 MPa. Scene addresses be-
485 long to their parsed source; they are not universal com-
486 ponent identifiers.

487 Truss FEM. For each pin-jointed member, the recov-
488 ered endpoints determine its length L and direction;
489 visible section dimensions determine area A , and the
490 material note supplies E . The axial element stiffness
491 is EA/L times the direction-cosine matrix. Assembly
492 and the recovered support conditions give $K_{ff}u_f = f_f$.
493 Member extension gives axial force and stress; support
494 reactions provide a global force and moment equilib-
495 rium check. The model assumes linear elasticity, small

displacement and nodal loading. The reported pass in
this implementation is a numerical equilibrium check,
not a stress-allowable or buckling certification.

Plate geometry. For circular holes in a rectangular
plate, the importer recovers the boundary, hole centres,
diameters, thickness and stated clearance criteria. It
checks boundary crossing, hole overlap, edge distance
and ligament, and reports net area and volume. For
two holes, the ligament is $\|c_i - c_j\| - r_i - r_j$; edge
clearance is the distance from a centre to the nearest
boundary minus its radius. These are manufacturing-
geometry checks; plate stress requires a separate con-
tinuum model and is not inferred from the geometric
pass.

510 6.2. Verified SVG edit examples

Five saved examples were replayed through the imple-
mentation for this revision; four reproduce the ear-
lier saved edited SVG trees, and the fifth is a clear-
ance violation selected from the audited dataset. Ta-
ble 6 lists their actual outputs. The source and edited
SVGs, requests, applied patches where applicable, and
re-imported verification records are retained with the
examples.

Full pipeline audit. The saved dataset audit streams
all 40,000 truss/plate SVG-edit rows through edit in-
terpretation, patch application, output re-import and
the applicable checker. All 40,000 match the stored
target tree and pass edit fidelity. All 20,008 truss
edits satisfy FEM equilibrium. Of 19,992 plate ed-
its, 19,293 meet the geometric criteria and 699 vio-
late edge-distance or ligament requirements. The 699
are successfully executed edits whose consequences are
flagged by the checker; discarding them as editing er-
rors would hide the distinction the workflow is designed
to expose. These synthetic, bounded requests demon-
strate pipeline consistency rather than learned gener-
alization.

533 6.3. Learned editing across other drawing families

A separate 7B patcher was trained on 8,000 lineage-
disjoint synthetic examples and evaluated on 100 bal-
anced held-out examples from five drawing families.
It produces valid patches in 100/100 cases, but exact
target-tree equality in 59/100. The family counts are
4/20 building plans, 20/20 circuits, 6/20 furniture ta-
bles, 9/20 mechanical parts and 20/20 water piping. A
drawing audit that allows numeric serialization differ-
ences finds 60/100 outputs with both the intended re-
covered parameters and internally consistent geometry.
This is a different adapter, dataset and metric from

Drawing family	Evaluated editor	Check on the output	Evidence and input contract
Circuits / pipes	Learned 1.5B/7B patches	Circuit solve; hydraulic solve	1,000 main and 400 hard edits; visible-geometry/label network importer
Axial trusses	Bounded deterministic edit	Axial FEM; equilibrium	20,008 audited edits; visible geometry, sections, loads and supports in a fixed grammar
Perforated plates	Bounded deterministic patch	Hole/edge/ligament geometry	19,992 audited edits; 699 faithful edits violate stated geometry rules
Planar frames	Saved Astra SVG edits	Geometry readback; frame FEM	Explicit physical model supplied; 21-member example, three analysis refusals in four attempts
Mechanical CAD	Saved Astra generation/edits	Task-specific shape checks	Ten completed hard-screen tasks pass; retained as controls

Table 5. Implemented checks and evaluated editors. Counts describe different protocols and are not pooled. FEM and geometry checks apply only to their stated physical or manufacturing assumptions.

Request	Edit mechanism	Recomputed quantity	Check on delivered SVG
Set all member depths to 40 mm	Tree patch	Peak stress: 11.861 $\rightarrow$ 5.930 MPa	Fidelity and FEM equilibrium pass
Double downward bridge load	Tree patch	Peak stress: 84.375 $\rightarrow$ 168.750 MPa	Fidelity and FEM equilibrium pass
Change bridge to eight panels	Structural patch	tree 21 $\rightarrow$ 29 members; peak stress 112.500 MPa	Connected output; FEM equilibrium passes
Enlarge six plate holes by 4 mm	Tree patch	Edge: 25 $\rightarrow$ 23 mm; ligament: 58 $\rightarrow$ 54 mm	Stated clearance rules pass
Enlarge four plate holes by 6 mm	Tree patch	Edge: 15 $\rightarrow$ 12 mm	Faithful edit; 15 mm edge rule fails

Table 6. Replayed engineering edits. These examples use the deterministic structural/mechanical route. The bridge topology example keeps load per loaded node constant, so total load changes with panel count. Stress is recomputed, but no material strength limit is tested by the truss equilibrium check.

the network editor in Tab. 4. Mechanical dimension notes match in all 20 examples while 11 corresponding drawings are geometrically inconsistent: checking annotations alone would overstate editing quality.

Frame and CAD analysis controls. The saved 21-member frame example in Fig. 7 passes geometry and dimension scoring in all four completed attempts; three explicitly decline the requested analysis, while one calculates within tolerance. Its separate FEM harness gives $u_x = 3.5119$ mm, $u_y = -0.1882$ mm and peak stress 17.6837 MPa after the edit. Ten hard-screen mechanical CAD outputs pass their geometric checkers, and six catalog-section structural designs pass SVG readback and their FEM constraints. These positive controls and the frame refusals establish checked examples under specific protocols; they do not establish a general CAD solver or an Astra failure on every engineering family.

7. Limitations and conclusion

Limits of the evidence. The drawing families are procedural and use bounded symbol and annotation grammars. The learned network editor, deterministic truss/plate editor, separate frame harness and cross-domain 7B patcher have different training and input contracts. Their results cannot be pooled or transferred across domains. The network importer rejects unsupported symbols and label placements. The structural importer covers idealised pin-jointed trusses; its FEM equilibrium pass checks numerical consistency, not buckling, fatigue, joints or material strength. Plate clearance checks supply no continuum stress analysis. The frame diagnostic receives an explicit physical model and is outside the network importer’s validated input domain.

The network Astra study uses one effort setting and one sample per task, with incomplete pipe coverage because API credits ran out. Its single wrong ver-

dict establishes a failure mode rather than a rate. The repeated frame refusal concerns one edited structure, with one correct answer among four attempts. The network editor's performance also depends on addressing: the saved 7B opaque-identifier control falls from 398/400 to 154/400 correct hard edits. The broader learned patcher obtains 60/100 consistent, intended drawings, so robust editing across arbitrary engineering sheets remains open. Numerical velocity scoring for Astra uses a 0.01 m/s absolute floor to avoid treating three-decimal rounding near zero as a substantive error; edit and verdict scores are unaffected.

Conclusion. The implemented workflow edits engineering SVGs and checks the design recovered from the result. Its strongest learned result is accurate editing of the supported circuit and pipe networks. Its structural/mechanical route demonstrates section and load edits, connected bridge topology patching, and detection of clearance violations in correctly patched plates. FEM, circuit, hydraulic and geometric checks make the consequences explicit. The resulting evidence identifies both successful edits and the engineering criteria that a requested revision fails, with each claim tied to the applicable model and delivered drawing.

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A. Supplementary material

A.1. Prompt given to Astra

System message:

You are an engineering drafting and analysis assistant. Use your own reasoning; no external tools are provided. Do not claim you ran a solver.

User message for a circuit task, verbatim up to the source SVG, which follows in full:

Instruction: Set R3 to 150 Ω .

Drawing conventions: straight lines are ideal wires. A wire that ends on another wire (a T) is connected to it; a filled dot also marks a connection; wires that cross without a dot are NOT connected. Rectangles are resistors and circles are ideal DC voltage sources; the '+' inside a source marks its positive terminal. Labels give component names and values.

Criteria (from the drawing notes): every resistor's dissipated power must not exceed the stated rating, and the magnitude of every source's current must not exceed the stated supply fuse rating.

Tasks:

1. Apply the instruction to the drawing, changing nothing else, and return the complete edited SVG.
2. Analyse the EDITED design and report whether it meets every criterion.

Answer with exactly two fenced blocks and nothing else: first ```svg containing the full edited SVG, then ```json containing {"verdict": "pass" or "fail", "violations": [names of every resistor or source that violates a criterion], "source_currents_A": {"V1": current delivered out of its + terminal, ...}, "max_resistor_power_W": {"name": "R?", "value": watts}}

Source SVG:

[complete source SVG]

For pipe networks the conventions paragraph and the JSON schema are replaced by:

Drawing conventions: heavy lines are pipes; filled dots are junctions. A pipe runs between junctions or reservoirs and may turn through 90-degree elbows; each elbow adds the minor-loss coefficient K given in the notes. 'P# L m ØD' gives a pipe's length (m) and internal diameter (mm); 'J# q=.. L/s z=.. m' gives the demand withdrawn at a junction and its elevation; 'R# H=.. m' is a reservoir's fixed total head.

Analysis model: steady incompressible water, density 998 kg/m³, dynamic viscosity 1.002e-3 Pa s, g = 9.80665 m/s²; Darcy-Weisbach head loss with the Churchill (1977) friction factor and the pipe roughness in the notes; tee and junction losses neglected; pressure head at a junction = hydraulic head minus elevation.

Criteria (from the drawing notes): every pipe velocity must not exceed the maximum velocity, and every junction pressure head must be at least the minimum pressure head.

JSON schema: {"verdict": "pass" or "fail", "violations": [names of every pipe or junction that violates a criterion], "pipe_velocities_m_s": {"P1": speed, ...}, "min_pressure_head_m": {"name": "J?", "value": metres}}

A.2. Worked Astra failure on the returned drawing

Figure 6 shows the actual source SVG and Astra's returned SVG for the one completed network answer with a substantive wrong verdict. The edit changes only P9's length from 290 to 340 m; the returned drawing

equals the target SVG exactly. The source is already below its 20 m pressure minimum at J4 (19.26 m), and re-solving the edited SVG gives 19.23 m. Astra instead reports 20.04 m and **pass**. Its reported velocity for P4 is 0.031 m/s versus 0.095 m/s from the edited drawing. These are analysis errors on a correctly edited artifact. The orange and red marks in the figure are presentation overlays added after the SVG was scored; they are not input to the verifier.

A.3. Separate structural-frame diagnostic

To test whether the focus on circuits and pipe networks hid other engineering-drawing difficulties, we re-audited saved Astra responses from a separate planar-frame study. The edit in Fig. 7 moves one column grid from 3300 to 4000 mm and increases the downward load at the top-right joint from 53 to 68 kN. The 21-member, three-storey frame has 27 degrees of static indeterminacy. Its prompt supplied the explicit physical graph as well as the source SVG, and no external solver. In three of four completed attempts on this same edit, Astra returned a drawing whose member geometry and dimensions pass the independent scorer, but explicitly marked displacement and stress as unavailable. In the fourth it gave values within 2 % of the frame FEM reference. Re-solving each returned drawing gives $u_x = 3.5119$ mm, $u_y = -0.1882$ mm and peak stress 17.6837 MPa. This is a repeated refusal to calculate, not a wrong numerical safety claim. It uses a separate planar-frame FEM harness; ReSolve does not verify these frames. Ten completed mechanical CAD hard-screen tasks and six functional frame-sizing controls also passed their respective saved-output audits, so neither provides an additional confirmed shape failure.

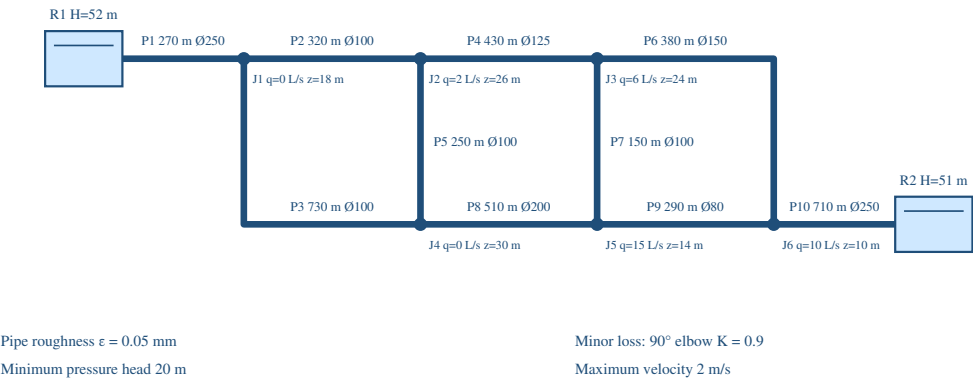
A.4. Results by edit type

Table 7 breaks end-to-end accuracy down by edit type. The one type on which Astra is below 100 % is the pipe-length change that produced its silent wrong verdict (Fig. 4). Positional addressing fails mostly on circuit edits, and id addressing is correct on every main-tier type at both model sizes.

A.5. Training stability

The first 7B training run, on an H100 NVL, diverged: the loss rose from 0.65 at step 20 to 3.0×10^4 at step 30 with a non-finite gradient norm, and the resulting adapter emits only "!" characters. Two controlled 50-step reruns isolate PyTorch's cuDNN scaled dot-product-attention kernel: with it, the gradient again became non-finite at step 30; without it, all steps were finite and the loss fell from 0.55 to 0.026. The reported

Water distribution network



(a) Source drawing: P9 is 290 m; J4 is already below its pressure limit.

Water distribution network



- **Astra edit: P9 290 to 340 m - drawing is correct**
Astra analysis: PASS; J4 = 20.04 m
- **Solve of returned SVG: FAIL; J4 = 19.23 m (min 20 m)**

(b) Astra’s correct SVG edit (orange P9) with the missed J4 violation (red).

Figure 6. Astra failure, shown on the complete engineering drawings. The two-reservoir, four-loop network is edited correctly, but Astra’s numerical answer declares it safe. The lower annotation contrasts that answer with an independent solve of the returned SVG. Original unannotated SVGs were used for scoring.

806
807
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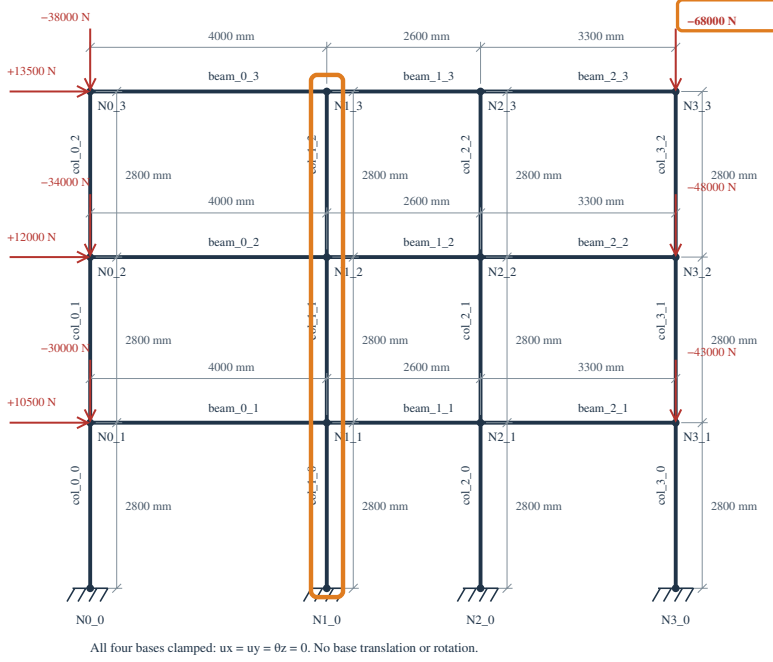
7B editor was retrained with that kernel disabled. The 1.5B editors, trained on an RTX PRO 6000, never used it. The diverged run is retained, marked void and ex-

cluded from every table. The trainer now stops at the first non-finite loss or gradient.

809
810

Three-bay three-storey frame elevation

Edited grid: $x = 4000 \text{ mm}$ · $N3_3$ vertical load: -68000 N · Undeformed member centrelines



SECTION SCHEDULE

Solid rectangles · $b \times h$ in mm

col_i_0	160 × 200
col_i_1	140 × 180
col_i_2	140 × 160
beam_i_1	140 × 260
beam_i_2	140 × 240
beam_i_3	140 × 220

Columns: $i = 0 \dots 3$; beams: $i = 0 \dots 2$.

h in plane; b out of plane.

All members: $E = 200000 \text{ N/mm}^2$.

RESULTS — EDITED FRAME

Probe $N3_3$ · $+ux$ right · $+uy$ upward

u_x : unavailable

u_y : unavailable

Peak normal stress: unavailable

Edited-frame stiffness analysis not evaluated.

Original drawing results have not been reused.

LIMIT CHECKS — NOT DETERMINED

$|u_y(N3_3)| \leq 12.0 \text{ mm}$: undetermined

Peak stress $\leq 180.0 \text{ MPa}$: undetermined

Stress definition, across all member ends:

$\max [|N|/A + |M|/(b^2/12)]$, $I = bh^3/12$

MODEL AND DRAWING NOTES

Rigid joints; linear elasticity and small displacements; axial and Euler–Bernoulli bending only. Loads act at the indicated nodes only.

Omitted: shear deformation, self-weight, buckling, out-of-plane behaviour and connection flexibility. Applied nodal moments are zero.

Dimensions are physical centreline lengths in mm. Load arrows are schematic, not force-scaled. All other nodes, sections and loads are unchanged.

Coordinate map: $X = 100 + (3/56)x$; $Y = 610 - (3/56)y$. Physical x and y in mm; drawing coordinates in SVG units.

Only the stated 2D frame is represented. Any future pass applies only to its named check, not to complete object or building safety.

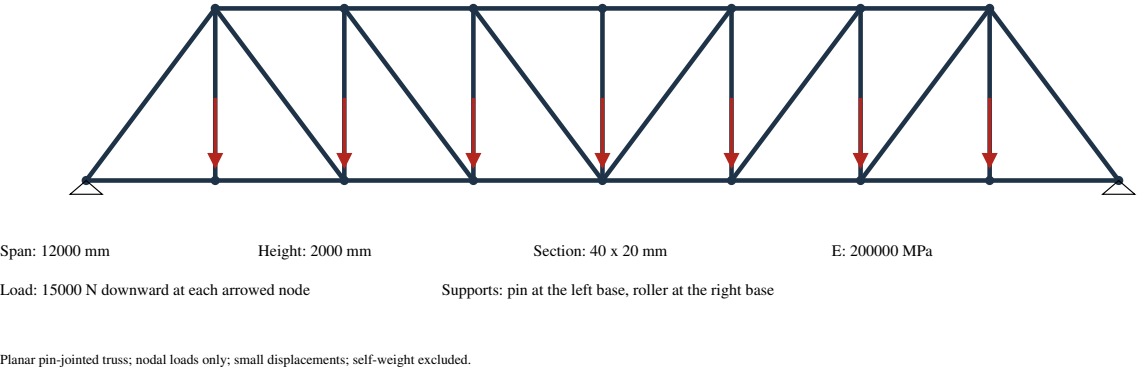
Figure 7. Astra’s returned SVG for a 21-member frame edit. The moved column grid and the increased load are highlighted in orange; the red box encloses the unavailable analysis. The editing is correct, but the requested stress and displacement analysis is absent in this attempt and two of three other completed attempts. The fourth attempt calculates correctly. Highlights are presentation overlays; the unmarked returned SVG was scored. This separate structural diagnostic is not included in ReSolve’s circuit and pipe-network benchmark.

Edit type	Astra	1.5B pos.	1.5B id	7B id
Circuits: component value	100 (10/10)	31.2 (39/125)	100 (125/125)	100 (125/125)
Circuits: open branch	100 (10/10)	41.6 (32/77)	100 (77/77)	100 (77/77)
Circuits: parallel branch	100 (10/10)	53.6 (67/125)	100 (125/125)	100 (125/125)
Circuits: polarity	100 (10/10)	29.2 (14/48)	100 (48/48)	100 (48/48)
Circuits: source value	100 (10/10)	52.8 (66/125)	100 (125/125)	100 (125/125)
Pipes: add pipe	100 (1/1)	100 (60/60)	100 (60/60)	100 (60/60)
Pipes: junction demand	100 (7/7)	66.4 (83/125)	100 (125/125)	100 (125/125)
Pipes: pipe diameter	100 (8/8)	99.2 (124/125)	100 (125/125)	100 (125/125)
Pipes: pipe length	83.3 (5/6)	100 (65/65)	100 (65/65)	100 (65/65)
Pipes: remove pipe	100 (7/7)	100 (125/125)	100 (125/125)	100 (125/125)

Table 7. End-to-end accuracy (edit, verdict and violations all correct), % (count). Astra on its completed main-tier tasks; ReSolve editors (positional or id addressing) on the full 1,000-edit main test set.

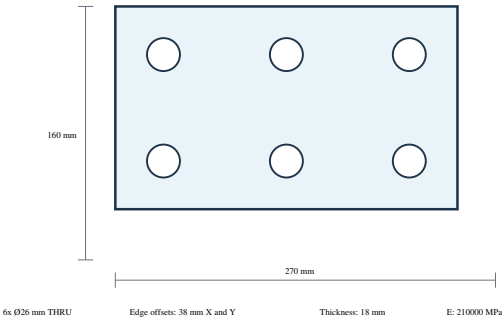
A.6. Worked structural and plate editing outputs

Figure 8 retains the visible engineering annotations of the replayed bridge and six-hole plate examples. The bridge uses a parsed structural-tree patch; the plate uses a tree patch. Both results are re-imported after editing. The exact source/output SVGs and verification JSON are saved in the accompanying example run, so each numerical statement can be checked against the produced artifact. The truss equilibrium pass is not a general structural safety certificate.



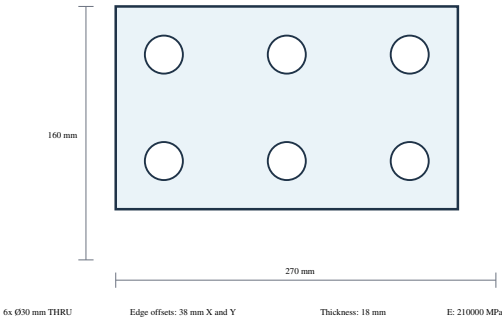
(a) Delivered 29-member bridge SVG; FEM: peak stress 112.50 MPa, maximum displacement 11.848 mm.

Untagged 6-hole mounting plate
Visible geometry and conventional dimensions; units mm



(b) Source six-hole plate.

Untagged 6-hole mounting plate
Visible geometry and conventional dimensions; units mm



(c) Hole diameters increased by 4 mm.

Figure 8. Actual SVG outputs of the verified editor. The bridge output preserves the stated span, height, section and per-node loading while changing connectivity. The plate’s recovered edge distance changes from 25 to 23 mm and ligament from 58 to 54 mm; both remain above the stated 15 and 20 mm minima. These are deterministic editor examples with FEM or geometry checks.